

Application of the transfer matrix approximation for wave propagation in a metafluid representing an acoustic black hole duct termination



Oriol Guasch*, Patricia Sánchez-Martín, Davide Ghilardi

GTM - Grup de recerca en Tecnologies Mèdia, La Salle, Universitat Ramon Llull, C/ Quatre Camins 30, Barcelona 08022, Catalonia, Spain

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ABSTRACT

The transfer matrix method has been proposed to analyze the acoustic black hole effect in duct terminations. The latter is achieved by placing a retarding waveguide structure inside the duct, which consists in a set of rings whose inner radii decrease to zero following a power law. The rings are separated by thin air cavities. If the number of ring-cavity ensembles is large enough, wave propagation inside the waveguide can be treated as a continuous problem. A governing differential equation can be derived for the acoustic black hole which intrinsically relies on assumptions from transfer matrix theory. However, no formal demonstration exists showing that the transfer matrix solution is consistent and formally tends to the solution of the continuous problem. Proving such consistency is the main goal of the paper and an original option has been adopted to this purpose. First, we prove the differential equation for the acoustic black hole is identical to the wave equation for a metafluid with a power-law varying density. Transfer matrices are then applied to solve wave propagation in a discretization of the metafluid into layers of constant density. It is shown that when the number of layers tends to infinity and their thicknesses to zero, the transfer matrix solution satisfies the metafluid equation and therefore the acoustic black hole one. The transfer matrices for the metafluid discrete layers take a particularly simple form, which is a great advantage. This work allows one to interpret the retarding waveguide structure as a particular realization of the metafluid.

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1. Introduction

The acoustic black hole (ABH) effect in mechanics originated with the pioneering work of Mironov [1], who showed that a flexural wave propagating in a power-law wedge at the end of a beam would progressively slow down concentrating its energy at the tip of the wedge. For a beam with zero end thickness, the wave would actually never reach the tip so no reflection could occur. However, a small truncation end thickness could ruin the ABH effect. It was not until several years later that it was proposed to place a viscoelastic damping layer at the wedge tip to dissipate energy and strongly reduce flexural wave propagation [2,3]. Since then, works on ABHs have experienced a major growth. Some have focused on analyzing the effects of imperfections at the truncation thickness [4] and studying the viscoelastic dissipation mechanisms [5,6]. Attempts to improve the ABH performance have included modifying the wedge geometry by means of extended platforms [7,8] or spi-

* Corresponding author.

E-mail address: oriol.guasch@salle.url.edu (O. Guasch).

ral rolling [9], exploiting energy transfer thanks to geometry nonlinearities [10], or substituting the viscoelastic layers with more efficient passive constrained viscoelastic layers [11]. Besides, circular and rectangular cuneate ABH indentations have been embedded in plates for vibration reduction and been analyzed with experimental [12,13], numerical [14] and semianalytical [15,16] methods. Likewise, ring-shaped ABHs have been recently proposed for vibration isolation in plates [17] and ABH arrays have been exploited for energy harvesting applications thanks to energy focalization [18]. ABH arrays also exhibit amazing properties in phononic plates, such as lensing and negative refraction and birefractive of flexural waves [19].

Despite most research to date has been devoted to structural ABHs, this paper deals with a completely different type of acoustic black holes. As Mironov and Pislyakov showed in 2002, an ABH effect can also be achieved for acoustic waves propagating in a duct [20]. They suggested the design of a retarding waveguide at the duct termination, which consisted of a set of rings separated by cavities. The rings were such that their inner radii decay to zero, following a power-law profile. Mironov and Pislyakov derived a one-dimensional differential equation to describe wave propagation inside the ABH, which should be valid in the case of a large number of cavity/ring ensembles. Once more, this idea did not receive attention until years later. In [21,22], linear and quadratic ABHs were built and experimental measurements were reported on their behavior, leaving some open questions concerning the efficiency of the absorption material in the ABHs. Shortly after, it was proposed to use the transfer matrix method (TMM) to analyze finite realizations of the ABHs, given that only a rather limited number of ring/cavity sets could be manufactured in practice [23]. Preliminary finite element simulations of duct ABH terminations were recently presented in [24]. Besides, the ABH retarding structure inspired some muffler designs in [25,26]. Beyond Mironov and Pislyakov one-dimensional ABH and relying on different principles, broadband two-dimensional ABHs have been also conceived by some authors. For instance, in [27,28] an annular shaped graded index metamaterial, which acts as an impedance matching layer between air and an absorber core, was constructed to dissipate waves coming from arbitrary directions. A similar design was suggested in [29] but using a second metamaterial at the core of the ABH structure. More recently, perfect, broadband absorption has been reported by means of subwavelength thickness panels with embedded Helmholtz resonators of graded dimensions [30].

In this work we will focus on some aspects concerning the TMM approach to simulate ABHs in duct terminations. The TMM method is widely employed in many other areas of physics which range from the vibrations of complex structures [31] to articulatory speech synthesis [32], to mention a couple. In most applications, it is assumed that the material properties and/or geometries are uniform over finite layers instead of experiencing gradual variations. However, no works can be found, to the best of the authors knowledge, concerning the limit behavior of the TMM when those layers tend to zero width, which is akin to consider their number grows to infinity. This mathematical issue constitutes the goal of this paper and it is of importance for wave propagation in ABH duct terminations for the following reason. If practical realizations of the ABHs can be correctly characterized by the TMM [23], then one should validate the TMM solution tends to the solution of the differential ABH equation proposed in [20], when the number of ring/cavity sets goes to infinity. In this work we address that problem and prove consistency of the TMM approach by showing that the TMM solution satisfies a finite difference discretization of the ABH continuous equation. Note that if we wanted to prove convergence of the TMM we should also prove the method is stable, but this is deemed out of the scope of the current work. Therefore, what it is actually demonstrated in this paper is that at least the TMM solution formally tends to that of the continuous problem.

To prove consistency of the TMM we have relied on the following strategy. First, we realize the differential ABH equation in [20] is the same one governing wave propagation in a metafluid, whose density increases according to a power-law while approaching the duct end wall. There is no general consensus for the definition of a metafluid, or a metamaterial, but according to [33] a good working one could be that of a fluid, or material, with on-demand effective properties and without the constraints imposed by what nature provides. This applies to our case. A fluid with a power-law density at a duct termination cannot be found in nature but we can manufacture it by placing a set of rings and cavities with power-law decreasing inner radii in a duct filled with a fluid of constant density. In other words, a real built-up ABH (see [21,22]) can be actually understood as a practical realization of a power-law density metafluid. The advantage of dealing with the latter is that the transfer matrices describing wave propagation in a discretization of the metafluid into uniform layers are simpler than those for the ring/cavities in [23], which facilitates demonstrating consistency of the TMM. By inserting the metafluid TMM solution in a finite difference discretization of the metafluid continuous equation we prove, by means of Taylor series expansions, that the TMM solution formally tends to the solution of the differential ABH equation when the number of metafluid layers goes to infinity and, accordingly, the layer thicknesses tend to zero. A preliminary version of this work dealing with the case of a linear power-law profile ABH was presented in [34]. The current work generalizes those results to ABHs of any order.

The paper is organized as follows. In Section 2, we review the derivation of the ABH equation in [20] and then introduce a one-dimensional wave equation for an inhomogeneous fluid with varying density. Both equations are shown to be equal if it is assumed the inhomogeneous fluid consists of a metafluid with a power-law density. We also review the smoothness condition for the proper functioning of an ABH. Besides, the TMM approximation for the metafluid is introduced in Section 3, whereas demonstrating consistency of the TMM solution is the topic of Section 4. Some simulations comparing the analytical and TMM reflection coefficients for linear and quadratic ABHs are shown in Section 5 for completion. The conclusions close the paper in Section 6.

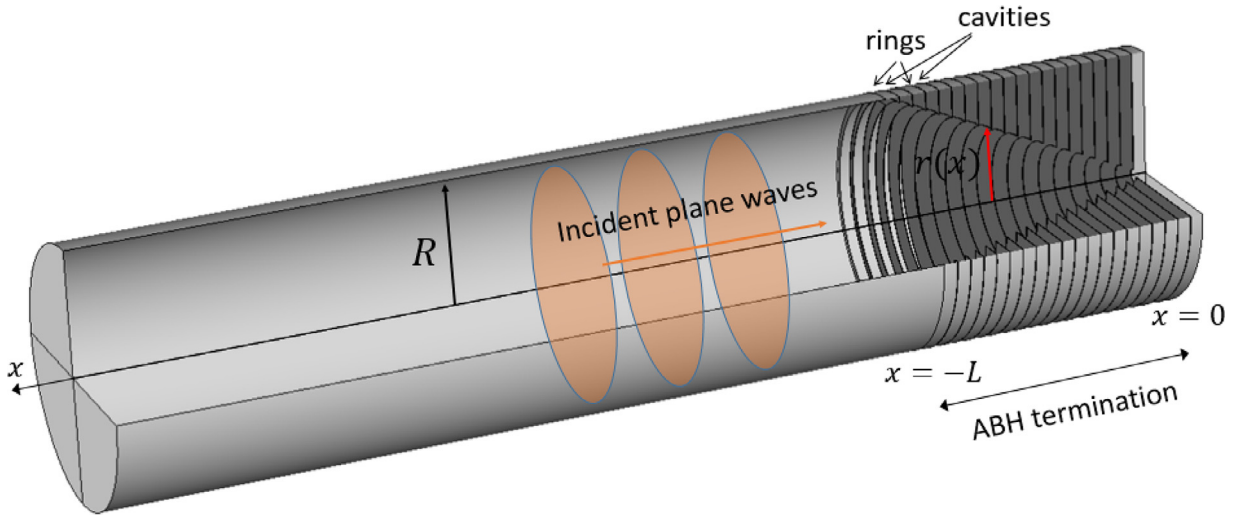


Fig. 1. Sketch of a cylindrical duct with an ABH termination. One quarter of the cylinder has been removed to illustrate the inner ABH structure of rings and cavities.

2. The ABH waveguide as a metafluid

2.1. Governing equation for the ABH in a duct termination

As explained in the Introduction, Mironov and Pislyakov [20] proposed a retarding structure to attain the ABH effect for plane wave propagation at a duct termination with circular cross-section. The structure comprises a length L and consists of a set of rigid rings, separated by air cavities, with inner radii $r(x)$ decreasing to zero following a power-law decay. Hereafter, x designates the longitudinal axis coordinate with origin at the duct termination (see Fig. 1), the ABH structure expanding from the entry at $x = -L$ to the end at $x = 0$. It was assumed in [20] that, in the limit of a very large number of ring-cavity ensembles, such system could be well approximated by the continuity and momentum equations describing isentropic wave propagation in an axisymmetric duct of varying cross section, $S(x) = \pi r^2(x)$, and wall admittance, $Y(x)$, to be determined below. These equations read,

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \rho u \frac{1}{S} \frac{\partial S}{\partial x} + \frac{2\rho Y p}{r} = 0, \quad (1a)$$

$$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \frac{\partial p}{\partial x} = 0, \quad (1b)$$

where $\rho(x, t)$ denotes the air density, $p(x, t)$ the air pressure and $u(x, t)$ the velocity, which only takes place in the x direction.

For small acoustic perturbations we may consider $\rho(x, t) = \rho_0 + \rho'(x, t)$, ρ_0 being the mean air density, $p(x, t) = p_0 + p'(x, t)$, p_0 standing for the ambient pressure and $u(x, t) = u'(x, t)$, which means the air is assumed to have zero mean velocity inside the duct. As we are considering wave propagation in air to be isentropic, the density and pressure fluctuations are related through $p'(x, t) = c_0^2 \rho'(x, t)$, c_0 denoting the sound speed. If we linearize Eqs. (1) and subtract the space derivative of (1b) from the time derivative of (1a), we get

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} + \frac{2\rho_0 Y}{r} \frac{\partial p'}{\partial t} - \frac{\partial^2 p'}{\partial x^2} + \left[S(x) \frac{\partial}{\partial x} \left(\frac{1}{S(x)} \right) \right] \frac{\partial p'}{\partial x} = 0, \quad (2)$$

which is nothing but a slight generalization of Webster's equation in the case of non-rigid duct walls. For harmonic time dependence, $p(x, t) = \hat{p}(x)e^{-i\omega t}$, one can express (2) in the frequency domain as,

$$-\frac{\partial^2 \hat{p}}{\partial x^2} - \left[k_0^2 + iZ_0 \frac{2Y}{r} k_0 \right] \hat{p} + \left[S(x) \frac{\partial}{\partial x} \left(\frac{1}{S(x)} \right) \right] \frac{\partial \hat{p}}{\partial x} = 0, \quad (3)$$

where as usual $k_0 = \omega/c_0$ represents the wavenumber, ω being the radial frequency, and $Z_0 = \rho_0 c_0$ denotes the air characteristic impedance.

As proposed in [20], let us first contemplate a power-law decay for the inner radius $r(x)$,

$$r(x) = (-1)^m \frac{R}{L^m} x^m, \quad (4)$$

with m being a positive integer known as the ABH order. Note that $r(-L) = R$ and $r(0) = 0$. In addition to (4), for achieving the ABH effect the wall admittance $Y(x)$ must have an appropriate dependence on $r(x)$. It was implicitly supposed in [20] that if one neglects the inner ring thicknesses and only considers those of the air cavities, a continuous wall admittance can be built as a limiting case of a TMM cavity lumped admittance (see [23]),

$$Y(x) = -i \frac{k_0}{Z_0} \frac{R^2 - r^2}{2r}. \quad (5)$$

Substituting (4) and (5) into (3), and considering $S = \pi r^2$, provides the ABH governing equation,

$$\frac{\partial^2 \hat{p}}{\partial x^2} + \frac{2m}{x} \frac{\partial \hat{p}}{\partial x} + \left(\frac{k_0 L^m}{x^m} \right)^2 \hat{p} = 0. \quad (6)$$

Waves satisfying (6) are such that their wavenumber, and therefore their wavelength, depend on the position inside the ABH. The wavelength diminishes as the wave approaches the duct termination and the propagation speed tends to zero, so it would take an infinite amount of time for the wave to reach the end of the duct. Consequently, no wave could be reflected from there and the structure of rings and cavities would act as an ABH. In practice, however, Eq. (6) presents a singularity at the origin and one needs to assume a certain imperfection in the retarding structure, which prevents a perfect ABH effect [20,23]. This is somewhat akin to what occurs with the truncation thickness in ABH indentations for beams and plates (see e.g. [3,5,11]). Moreover, the power-law decay of the ring inner radii cannot take place for an arbitrary length, L , and a certain smoothness condition becomes necessary to avoid reflections at the various ABH sections. Again, this is analogous to what occurs for beam and plate ABHs [1,11,14]. We will review the implications of the smoothness condition in Section 2.4 below.

2.2. Governing equation for plane waves in a duct filled with a non-homogeneous fluid

Let us next address the apparently very different topic of plane wave propagation inside a duct filled with a non-homogeneous fluid with varying density. The one-dimensional continuity and momentum equations of the fluid read

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0, \quad (7a)$$

$$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \frac{\partial p}{\partial x} = 0, \quad (7b)$$

with $\rho(x, t)$, $p(x, t)$ and $u(x, t)$ respectively representing the air density, pressure and velocity, as in (1). The state equation is given by

$$\frac{\partial p}{\partial t} - \frac{B_s}{\rho} \left(\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} \right) = 0, \quad (8)$$

where we assume the flow compressibility $B_s(x)$ to only depend on the position x . Combining (7a) and (8) we obtain

$$\frac{\partial p}{\partial t} + B_s \frac{\partial u}{\partial x} = 0. \quad (9)$$

Assuming again small perturbations $\rho(x, t) = \rho_s(x) + \rho'(x, t)$, $p(x, t) = p_0 + p'(x, t)$ and $u(x, t) = u'(x, t)$ (no mean flow) in (7b) and (9), leaves, after linearization,

$$\frac{\partial p'}{\partial t} + B_s(x) \frac{\partial u'}{\partial x} = 0, \quad (10a)$$

$$\rho_s(x) \frac{\partial u'}{\partial t} + \frac{\partial p'}{\partial x} = 0. \quad (10b)$$

Again, it becomes feasible to build a wave equation from (10a) and (10b) by taking, as usual, the partial time derivative of (10a) minus the spatial derivative of $B_s(x)$ times (10b). After some straightforward manipulations, this produces

$$\rho_s(x) \frac{\partial^2 p'}{\partial t^2} - B_s(x) \frac{\partial^2 p'}{\partial x^2} + \left[\frac{B_s(x)}{\rho_s(x)} \frac{\partial \rho_s(x)}{\partial x} \right] \frac{\partial p'}{\partial x} = 0,$$

which, defining the space dependent wave speed $c_s(x) := [B_s(x)/\rho_s(x)]^{1/2}$ can be rewritten as

$$\frac{1}{c_s^2(x)} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x^2} + \left[\frac{1}{\rho_s(x)} \frac{\partial \rho_s(x)}{\partial x} \right] \frac{\partial p'}{\partial x} = 0. \quad (11)$$

Eq. (11) describes one-dimensional wave propagation in a non-homogeneous medium (see e.g., [35,36]) and it can be expressed in the frequency domain as

$$-\frac{\partial^2 \hat{p}}{\partial x^2} - k_s^2(x) \hat{p} + \left[\frac{1}{\rho_s(x)} \frac{\partial \rho_s(x)}{\partial x} \right] \frac{\partial \hat{p}}{\partial x} = 0, \quad (12)$$

with $k_s(x) \equiv \omega/c_s(x)$.

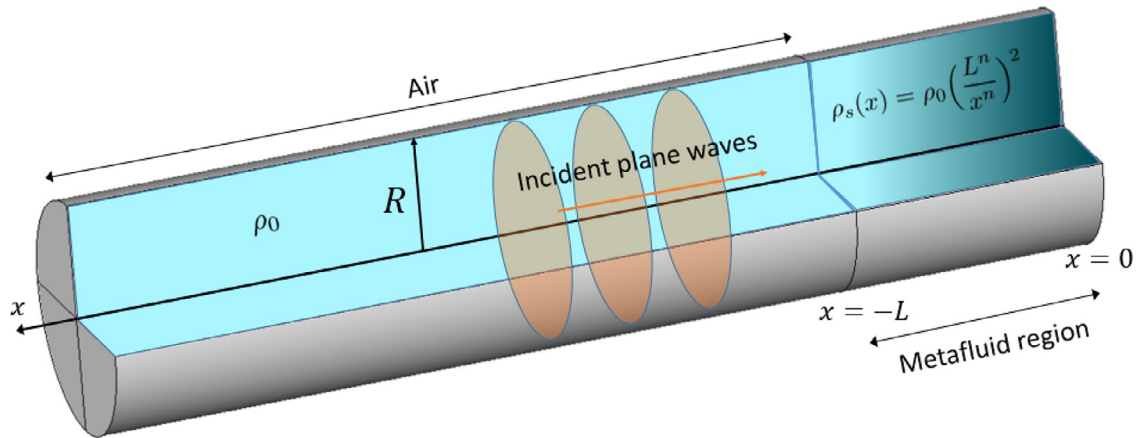


Fig. 2. Metafluid at the duct termination whose governing equation is analogous to that of the ABH in Fig. 1.

2.3. The ABH as a power-law density metafluid

We shall next consider that the flow filling the duct consists of a metafluid (non-existent in nature), the density of which augments when approaching the duct end following the power-law profile (see Fig. 2),

$$\rho_s(x) = \rho_0 \left(\frac{L^m}{x^m} \right)^2. \quad (13)$$

Note that the density is singular at the origin. Besides, suppose a constant metafluid compressibility $B_s(x) \equiv B_0$, leading to a sound speed $c_s(x) = \sqrt{B_0/\rho_s(x)}$, and therefore to a squared wavenumber,

$$k_s^2(x) = \omega^2 \frac{\rho_s(x)}{B_0} = \left(\frac{k_0 L^m}{x^m} \right)^2, \quad (14)$$

where use has been made of (13) in the last equality of (14), together with $c_0 = \sqrt{B_0/\rho_0}$. If we next insert (13) and (14) into (12), we recover the ABH Eq. (6), namely,

$$\frac{\partial^2 \hat{p}}{\partial x^2} + \frac{2m}{x} \frac{\partial \hat{p}}{\partial x} + \left(\frac{k_0 L^m}{x^m} \right)^2 \hat{p} = 0. \quad (15)$$

This result establishes that the equation governing acoustic wave propagation inside a duct, whose inner radius decreases to zero following the power-law (4) and whose wall impedance is given by (5), is exactly the same equation describing how waves would propagate in a metafluid having the power-law density (13). This physical analogy will be a key element to prove the consistency of the TMM solution in subsequent sections.

On the other hand, we can introduce a change of variables to express (15) as a Helmholtz equation with spatially varying wavenumber. If one considers $\hat{p} = \rho_s^{1/2} \phi$ in the non-homogeneous fluid Eq. (12), this transforms to [35],

$$\frac{\partial^2 \phi}{\partial x^2} + \kappa^2(x) \phi = 0, \quad (16)$$

with wavenumber

$$\kappa^2(x) = k_s^2(x) - \frac{3}{4\rho_s^2} \left(\frac{\partial \rho_s}{\partial x} \right)^2 + \frac{1}{2\rho_s} \frac{\partial^2 \rho_s}{\partial x^2}. \quad (17)$$

Substituting (13) and (14) into (17) provides

$$\kappa^2(x) = \left(\frac{k_0 L^m}{x^m} \right)^2 + \frac{m-m^2}{x^2}. \quad (18)$$

Therefore, solving the ABH and metafluid Eq. (15) is akin to solving the Helmholtz equation (16) for ϕ , with spatially varying wavenumber (18), and then recovering the acoustic pressure from $\hat{p} = \rho_s^{1/2} \phi$. This will be another key issue to demonstrate consistency of the TMM solution.

2.4. The smoothness condition

There is a further aspect to be considered for the proper functioning of a duct ABH waveguide. If the ABH length L is too short and the power law decay too steep, a wave entering the ABH will not perceive a smooth matching of impedance and

will be partially reflected back, making the ABH waveguide useless. Mironov was already aware of this fact in his pioneering work on ABH wedges for beams and plates and proposed that the variation of the acoustic wavenumber needs to be small over a distance of the order of a wavelength. Such condition can be expressed as [1],

$$\frac{1}{\kappa^2} \frac{d\kappa}{dx} \ll 1. \quad (19)$$

Inserting (18) into (19), the smoothness condition implies,

$$\frac{k_0^2 L^{2m}}{x^{2m}} \gg \frac{m^2}{x^2}. \quad (20)$$

In the most critical case of $x = L$, the condition simply states that $k_0 L \gg m$, which reflects the logical fact that the higher the order m of the ABH, the larger the termination length L must be to get a good impedance matching, because the power-law profile gets steeper.

In the case of acoustic wave propagation in an ABH duct termination, Mironov and Pisyakov assumed that the smoothness condition was in fact dictated by the conditions required for applying the WKB approximation to the problem [20]. Those are summarized in the fulfillment of

$$|\kappa| \gg \frac{1}{x}. \quad (21)$$

From (18) (see also Eq. (11) in [20]), the wave number κ is given by

$$\kappa(x) = (-1)^m \frac{L^m}{x^m} \sqrt{k_0^2 - \frac{m^2 x^{2m-2}}{L^{2m}}},$$

which once inserted in (21) yields

$$\frac{k_0^2 L^{2m}}{x^{2m}} \gg \frac{m^2 + 1}{x^2}. \quad (22)$$

This is a slightly more restrictive condition than (20). For the consistency demonstration, however, a weaker requirement than (22) or (20) suffices, namely,

$$\frac{k_0^2 L^{2m}}{x^{2m}} \gg \frac{m^2 - m}{x^2}, \quad (23)$$

which will allow us to approximate the wavenumber in the exponential functions of Section 4 below as,

$$\kappa = \sqrt{\frac{k_0^2 L^{2m}}{x^{2m}} - \frac{m - m^2}{x^2}} \approx \sqrt{\frac{k_0^2 L^{2m}}{x^{2m}}} = \frac{k_0 L^m}{x^m}.$$

3. The TMM for multilayered media

So far, we have been dealing with continuous problems driven by differential equations. In fact, and as said before, the ABH Eq. (6) can be understood as the result of the homogenization process of an ABH waveguide consisting of a finite number of ring plus cavity ensembles. In other words, such ABH finite structures (e.g. those built in [21,22] or those analyzed in [23]) can be viewed as particular practical realizations of the metafluid in the previous section.

As mentioned in the Introduction, the TMM is commonly used in many areas of physics to analyze wave propagation through multilayered media. Consequently, the TMM is an optimum candidate to analyze the performance of built-up ABH waveguides. In most applications of the TMM, however, the propagation media properties are constant and clearly distinct over regions of finite extension, and do not exhibit a gradual, continuous variation. Although one may guess a homogenization process is feasible and that, for example, the TMM solution to an ABH should satisfy a differential equation like (15) in the limit of an increasing number of rings/cavity sets, no rigorous proof has confirmed this hypothesis, as far as the authors know. As explained before, this constitutes the main purpose of the present work.

We have also justified in the Introduction that applying the TMM to the metafluid, instead of resorting to a more complex option, will facilitate demonstrating consistency. To this purpose we can rely on the developments in [37], intended to describe wave propagation in media of non-uniform density and wave speed. The intention is to solve the Helmholtz equation (16) with varying wavenumber (18) using the TMM. The region of inhomogeneous density, which expands from $x = -L$ to $x = 0$, is divided into N layers of identical width L/N (see Fig. 3). The physical parameters inside every layer are assumed to be constant. The arbitrary n -th layer, centered at x_n , will thus have constant density ρ_n and speed of sound c_n , and a characteristic impedance $z_n = \rho_n c_n$. In what follows, let us designate the approximate value of any function $g(x)$ at x_n as $g_n \approx g(x_n)$.

Given that inside every n -th layer the parameters are constant, the propagating waves would satisfy there a standard Helmholtz equation with fixed wavenumber κ_n . The solution in the layer would combine planar incident waves, ϕ_n^i , and reflected waves, ϕ_n^r , with expressions,

$$\phi_n^i = A_n e^{i(\omega t - \kappa_n x_n)}, \quad \phi_n^r = B_n e^{i(\omega t + \kappa_n x_n)}, \quad (24)$$

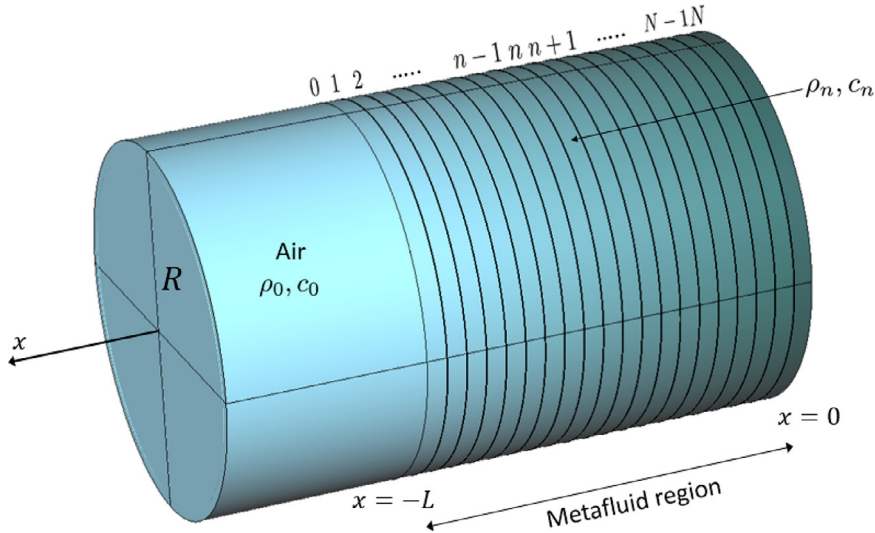


Fig. 3. TMM approach to describe wave propagation inside the metafluid.

and corresponding velocities ϕ_n^i/z_n and $-\phi_n^r/z_n$. One can relate the amplitudes $(A_n, B_n)^T$ at the n -th layer to those at the $(n+1)$ -th layer, $(A_{n+1}, B_{n+1})^T$, by imposing the continuity of ϕ^i , ϕ^r and their velocities at the boundary [37]. The relation takes place through a transfer matrix $\mathbf{T}_n$, such that,

$$\begin{pmatrix} A_n \\ B_n \end{pmatrix} = \mathbf{T}_n \begin{pmatrix} A_{n+1} \\ B_{n+1} \end{pmatrix}, \quad (25)$$

with

$$\mathbf{T}_n = \frac{1}{2z_{n+1}} \begin{pmatrix} (z_{n+1} + z_n)e^{-i(\kappa_{n+1}-\kappa_n)x_n} & (z_{n+1} - z_n)e^{i(\kappa_{n+1}+\kappa_n)x_n} \\ (z_{n+1} - z_n)e^{-i(\kappa_{n+1}+\kappa_n)x_n} & (z_{n+1} + z_n)e^{i(\kappa_{n+1}-\kappa_n)x_n} \end{pmatrix}.$$

Furthermore, we can define a matrix $\tilde{\mathbf{T}}_{(n,n+m)}$ to relate amplitudes between arbitrary layers n and $n+m$ as,

$$\tilde{\mathbf{T}}_{(n,n+m)} \equiv \prod_{i=0}^{m-1} \mathbf{T}_{n+i},$$

from which follows

$$\begin{pmatrix} A_n \\ B_n \end{pmatrix} = \tilde{\mathbf{T}}_{(n,n+m)} \begin{pmatrix} A_{n+m} \\ B_{n+m} \end{pmatrix}.$$

As a particular case, we may link the amplitudes $(A_N, B_N)^T$ at the duct termination, $x=0$, with the amplitudes $(A_0, B_0)^T$ at the entry, $x=-L$, of the ABH waveguide, through

$$\begin{pmatrix} A_0 \\ B_0 \end{pmatrix} = \tilde{\mathbf{T}}_{(0,N)} \begin{pmatrix} A_N \\ B_N \end{pmatrix}, \quad \begin{pmatrix} A_N \\ B_N \end{pmatrix} = \tilde{\mathbf{T}}_{(0,N)}^{-1} \begin{pmatrix} A_0 \\ B_0 \end{pmatrix}. \quad (26)$$

4. Consistency of the TMM solution

With all the above developments, we are now in disposition to demonstrate that the TMM solution formally tends to the solution of the ABH governing differential equation. As previously said, our strategy will consist in proving this for the TMM solution of the metafluid, whose governing equation coincides with (6).

Let us first introduce some notation. Omitting the factor $e^{i\omega t}$ in (24), we define the vector $\tilde{\phi}_n$, for the n th layer as

$$\tilde{\phi}_n = \begin{pmatrix} \phi_n^i \\ \phi_n^r \end{pmatrix} = \begin{pmatrix} A_n e^{-i\kappa_n x_n} \\ B_n e^{i\kappa_n x_n} \end{pmatrix} = \begin{pmatrix} e^{-i\kappa_n x_n} & 0 \\ 0 & e^{i\kappa_n x_n} \end{pmatrix} \begin{pmatrix} A_n \\ B_n \end{pmatrix} \equiv \mathbf{D}_n \begin{pmatrix} A_n \\ B_n \end{pmatrix}. \quad (27)$$

Combining (25) with (27) provides the following relation between vectors $\tilde{\phi}_n$ and $\tilde{\phi}_{n+1}$,

$$\tilde{\phi}_n = \begin{pmatrix} \phi_n^i \\ \phi_n^r \end{pmatrix} = \mathbf{D}_n \mathbf{T}_n \mathbf{D}_{n+1}^{-1} \begin{pmatrix} \phi_{n+1}^i \\ \phi_{n+1}^r \end{pmatrix} = \mathbf{D}_n \mathbf{T}_n \mathbf{D}_{n+1}^{-1} \tilde{\phi}_{n+1}. \quad (28)$$

Given that the solution ϕ_n at the n -th layer will consist of the summation of the incident and reflected waves, Eq. (28) allows one to write it as,

$$\phi_n = \phi_n^i + \phi_n^r = (1, 1) \begin{pmatrix} \phi_n^i \\ \phi_n^r \end{pmatrix} = (1, 1) \tilde{\phi}_n = (1, 1) \mathbf{D}_n \mathbf{T}_n \mathbf{D}_{n+1}^{-1} \tilde{\phi}_{n+1}. \quad (29)$$

Next, consider the following second order finite difference approximation of the Helmholtz Eq. (16) at point x_{n-1} ,

$$\frac{\phi_n - 2\phi_{n-1} + \phi_{n-2}}{(\Delta x)^2} + \kappa_{n-1}^2 \phi_{n-1} = 0, \quad (30)$$

in which the spacing Δx is taken to coincide with the layer width L/N . Using (29), the finite difference scheme for the second order derivative of ϕ at x_{n-1} in (30) can be expressed as,

$$\left. \frac{\partial^2 \phi}{\partial x^2} \right|_{n-1} \approx \frac{\phi_n - 2\phi_{n-1} + \phi_{n-2}}{(\Delta x)^2} = (1, 1) \frac{\tilde{\phi}_n - 2\tilde{\phi}_{n-1} + \tilde{\phi}_{n-2}}{(\Delta x)^2}, \quad (31)$$

while from (28) we get

$$\tilde{\phi}_{n-1} = \mathbf{D}_{n-1} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \tilde{\phi}_n, \quad (32)$$

and

$$\tilde{\phi}_{n-2} = \mathbf{D}_{n-2} \mathbf{T}_{n-2} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \tilde{\phi}_n, \quad (33)$$

which once inserted in (31) yield

$$\left. \frac{\partial^2 \phi}{\partial x^2} \right|_{n-1} \approx \frac{1}{(\Delta x)^2} (1, 1) (\mathbf{I} - 2\mathbf{D}_{n-1} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} + \mathbf{D}_{n-2} \mathbf{T}_{n-2} \mathbf{T}_{n-1} \mathbf{D}_n^{-1}) \tilde{\phi}_n, \quad (34)$$

with $\mathbf{I}$ standing for the identity matrix.

On the other hand, the second term of the discretized Helmholtz Eq. (30) becomes,

$$\begin{aligned} \kappa^2(x_{n-1}) \phi_{n-1} &= \left[\left(\frac{k_0 L^m}{x_{n-1}^m} \right)^2 + \frac{m - m^2}{x_{n-1}^2} \right] (1, 1) \tilde{\phi}_{n-1} \\ &= \left[\left(\frac{k_0 L^m}{x_{n-1}^m} \right)^2 + \frac{m - m^2}{x_{n-1}^2} \right] (1, 1) \mathbf{D}_{n-1} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \tilde{\phi}_n. \end{aligned} \quad (35)$$

As is well known, the finite difference solution to the Helmholtz Eq. (30) tends to the analytical one (16) for $\Delta x \rightarrow 0$. Therefore, in order to verify that the TMM solution also formally tends to the solution of the continuous problem we need to substitute (34) and (35) into (30), and check if the following limit is satisfied,

$$\lim_{\Delta x \rightarrow 0} \left\{ (1, 1) \left[\frac{1}{(\Delta x)^2} \mathbf{I} + \left(\kappa_{n-1}^2 - \frac{2}{(\Delta x)^2} \right) \mathbf{D}_{n-1} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} + \frac{1}{(\Delta x)^2} \mathbf{D}_{n-2} \mathbf{T}_{n-2} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \right] \tilde{\phi}_n \right\} = 0. \quad (36)$$

The limit (36) can be proved by means of a Taylor series expansion of all matrix elements involved in the expression. For instance, the elements in matrix $\mathbf{T}_{n-1}$ can be expanded as

$$T_{n-1}^{1,1} = 1 + \frac{2imk_0 L^m x_n - mx_n^m}{2x_n^{m+1}} \Delta x + \frac{-2m^2 k_0^2 L^{2m} x_n^2 - 2imk_0 L^m x_n^{m+1} + (m^2 - m)x_n^{2m}}{4x_n^{2m+2}} (\Delta x)^2 \quad (37a)$$

$$T_{n-1}^{1,2} = -\frac{me^{\frac{2iS_n}{x_n}}}{2x_n} \Delta x + \frac{(4im - 2im^2)k_0 L^m x_n e^{\frac{2iS_n}{x_n}} + x_n^m (m^2 - m)e^{\frac{2iS_n}{x_n}}}{4x_n^{m+2}} (\Delta x)^2 \quad (37b)$$

$$T_{n-1}^{2,1} = -\frac{me^{-\frac{2iS_n}{x_n}}}{2x_n} \Delta x + \frac{(2im^2 - 4im)k_0 L^m x_n e^{-\frac{2iS_n}{x_n}} + x_n^m (m^2 - m)e^{-\frac{2iS_n}{x_n}}}{4x_n^{m+2}} (\Delta x)^2 \quad (37c)$$

$$T_{n-1}^{2,2} = 1 - \frac{2imk_0 L^m x_n + mx_n^m}{2x_n^{m+1}} \Delta x + \frac{-2m^2 k_0^2 L^{2m} x_n^2 + 2imk_0 L^m x_n^{m+1} + (m^2 - m)x_n^{2m}}{4x_n^{2m+2}} (\Delta x)^2 \quad (37d)$$

while those of $\mathbf{T}_{n-2}$ become,

$$T_{n-2}^{1,1} = 1 + \frac{2imk_0 L^m x_n - mx_n^m}{2x_n^{m+1}} \Delta x + \frac{-2m^2 k_0^2 L^{2m} x_n^2 + (4im^2 - 2im)k_0 L^m x_n^{m+1} + (m^2 - 3m)x_n^{2m}}{4x_n^{2m+2}} (\Delta x)^2 \quad (38a)$$

$$T_{n-2}^{1,2} = -\frac{me^{\frac{2iS_n}{x_n}}}{2x_n} \Delta x + \frac{(8im - 6im^2)k_0 L^m x_n e^{\frac{2iS_n}{x_n}} + x_n^m (m^2 - 3m)e^{\frac{2iS_n}{x_n}}}{4x_n^{m+2}} (\Delta x)^2 \quad (38b)$$

$$T_{n-2}^{2,1} = -\frac{me^{-\frac{2iS_n}{x_n}}}{2x_n} \Delta x + \frac{(6im^2 - 8im)k_0 L^m x_n e^{-\frac{2iS_n}{x_n}} + x_n^m (m^2 - 3m) e^{-\frac{2iS_n}{x_n}}}{4x_n^{m+2}} (\Delta x)^2 \quad (38c)$$

$$T_{n-2}^{2,2} = 1 - \frac{2imk_0 L^m x_n + mx_n^m}{2x_n^{m+1}} \Delta x + \frac{-2m^2 k_0^2 L^{2m} x_n^2 + (2im - 4im^2)k_0 L^m x_n^{m+1} + (m^2 - 3m)x_n^{2m}}{4x_n^{2m+2}} (\Delta x)^2. \quad (38d)$$

In the matrix entries above, the parameters S_n can be simplified resorting to the smoothness condition (23), namely $S_n \equiv \sqrt{k_0^2 L^{2m} + (m - m^2)x_n^{2m-2}} \approx \sqrt{k_0^2 L^{2m}} = k_0 L^m$. For the linear case addressed in [34] no approximation is needed because $m = 1$ and therefore $m - m^2 = 0$.

Inserting (37) and (38), as well as the corresponding expansions for $\mathbf{D}_{n-1}$, $\mathbf{D}_{n-2}$ and $\mathbf{D}_n^{-1}$ into (36) to check its fulfillment, involves a considerable amount of lengthy algebraic manipulations. Those have been mostly handmade and checked with the PARI-GP mathematical software [38].

The final step of the demonstration consists in reverting the change of variable $\phi = \rho_s^{-1/2} \hat{p}$ and inspecting whether the TMM solution for the acoustic pressure $\hat{p}_n$ satisfies, in the limit, the ABH Eq. (15). Notice that the pressure at the n -th layer is given by

$$\hat{p}_n = \hat{p}_n^i + \hat{p}_n^r = \rho_s^{1/2}(x_n)(\phi_n^i + \phi_n^r) = \rho_0^{1/2} \frac{L^m}{x_n^m} (\phi_n^i + \phi_n^r),$$

whereas the matrix expressions analogous to (28), (32) and (33) for the acoustic pressure read,

$$\tilde{\mathbf{P}}_n = \begin{pmatrix} \hat{p}_n^i \\ \hat{p}_n^r \end{pmatrix} = \rho_s^{1/2}(x_n) \begin{pmatrix} \phi_n^i \\ \phi_n^r \end{pmatrix} = \frac{\rho_s^{1/2}(x_n)}{\rho_s^{1/2}(x_{n+1})} \mathbf{D}_n \mathbf{T}_n \mathbf{D}_{n+1}^{-1} \begin{pmatrix} \hat{p}_{n+1}^i \\ \hat{p}_{n+1}^r \end{pmatrix} = \frac{x_n^m}{x_{n+1}^m} \mathbf{D}_n \mathbf{T}_n \mathbf{D}_{n+1}^{-1} \tilde{\mathbf{P}}_{n+1},$$

$$\tilde{\mathbf{P}}_{n-1} = \frac{x_n^m}{x_{n-1}^m} \mathbf{D}_{n-1} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \tilde{\mathbf{P}}_n,$$

and

$$\tilde{\mathbf{P}}_{n-2} = \frac{x_{n-1}^m}{x_{n-2}^m} \mathbf{D}_{n-2} \mathbf{T}_{n-2} \mathbf{D}_{n-1}^{-1} \tilde{\mathbf{P}}_{n-1} = \frac{x_{n-1}^m}{x_{n-2}^m} \mathbf{D}_{n-2} \mathbf{T}_{n-2} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \tilde{\mathbf{P}}_n.$$

A plausible second order finite difference scheme for (15) at point x_{n-1} is

$$\frac{\hat{p}_n - 2\hat{p}_{n-1} + \hat{p}_{n-2}}{(\Delta x)^2} + \frac{2m}{x_{n-1}} \frac{\hat{p}_n - \hat{p}_{n-2}}{2\Delta x} + \left(\frac{k_0 L^m}{x_{n-1}^m} \right)^2 \hat{p}_{n-1} = 0. \quad (39)$$

The first and third terms of (39) can be treated analogously to those of (30). As regards the second one, which involves the first order derivative of the pressure at x_{n-1} , we get

$$\left. \frac{\partial \hat{p}}{\partial x} \right|_{n-1} \approx \frac{\hat{p}_n - \hat{p}_{n-2}}{2\Delta x} = \frac{1}{2\Delta x} (1, 1) \mathbf{D}_{n-2} \left(\mathbf{D}_{n-2}^{-1} \mathbf{D}_n - \frac{x_n^m}{x_{n-2}^m} \mathbf{T}_{n-2} \mathbf{T}_{n-1} \right) \mathbf{D}_n^{-1} \tilde{\mathbf{P}}_n.$$

To prove that $\hat{p}_n$ tends to the continuous pressure $\hat{p}$ of the ABH equation when $\Delta x \rightarrow 0$, the following limit needs to hold

$$\lim_{\Delta x \rightarrow 0} \left\{ (1, 1) \left[\frac{1}{(\Delta x)^2} \left(\mathbf{I} - 2 \frac{x_n^m}{x_{n-1}^m} \mathbf{D}_{n-1} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} + \frac{x_n^m}{x_{n-2}^m} \mathbf{D}_{n-2} \mathbf{T}_{n-2} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \right) + \frac{m}{x_{n-1} \Delta x} \left(\mathbf{I} - \frac{x_n^m}{x_{n-2}^m} \mathbf{D}_{n-2} \mathbf{T}_{n-2} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \right) + \frac{k_0^2 L^{2m}}{x_{n-1}^{2m}} \left(\frac{x_n^m}{x_{n-1}^m} \mathbf{D}_{n-1} \mathbf{T}_{n-1} \mathbf{D}_n^{-1} \right) \right] \tilde{\mathbf{P}}_n = 0. \right.$$

Once more, one can verify the limit is satisfied by means of Taylor series expansions of the entries in matrices $\mathbf{T}_{n-1}$, $\mathbf{T}_{n-1}$, $\mathbf{D}_{n-1}$, $\mathbf{D}_{n-2}$ and $\mathbf{D}_n^{-1}$. The involved algebraic developments can be asserted again with the help of the mathematical software package PARI-GP. This concludes the demonstration.

5. Numerical results

5.1. The ABH reflection coefficient

In the precedent section, it has been shown how the TMM solution to the ABH will approach the solution of the ABH differential equation. This would suffice for the purpose of this work though, in the sequel, we will present some numerical simulations for completeness.

Given that the main feature of an ABH concerns its ability to absorb incident waves, an ABH is usually characterized by means of the reflection coefficient at its entrance $x = -L$ (see e.g., [2,6,20,23] among others). The reflection coefficient can

be computed once the exact solution to the ABH governing Eq. (6), and hence to its metafluid analogue, Eq. (15), is known. Those were respectively derived for the cases of a linear ABH ($m = 1$) and a quadratic ABH ($m = 2$) in [20] and [23]. The solution to equations (6) and (15) for $m = 1$ becomes

$$p(x) = C_+ e^{\alpha_+ \ln x} + C_- e^{\alpha_- \ln x} \quad (40)$$

with C_+ and C_- standing for real constants and

$$\alpha_{\pm} = \frac{1}{2} \left[-1 \pm \sqrt{1 - (2k_0 L)^2} \right].$$

Besides, the solution for $m = 2$ is given by

$$p(x) = \frac{C_+}{x} \sqrt{(k_0 L^2)^2 + x^2} \exp \left\{ i \left[\frac{k_0 L^2}{x} - \arctan \left(\frac{k_0 L^2}{x} \right) \right] \right\} + \frac{C_-}{x} \sqrt{(k_0 L^2)^2 + x^2} \exp \left\{ -i \left[\frac{k_0 L^2}{x} - \arctan \left(\frac{k_0 L^2}{x} \right) \right] \right\}. \quad (41)$$

Using (40) and (41) to impose pressure and velocity continuity at the ABH entrance and prescribing a boundary admittance Y_0 at the termination $x = 0$, it is possible to deduce the linear and quadratic reflection coefficients at $x = -L$, namely $\mathcal{R}_L^{\text{lin}}$ and $\mathcal{R}_L^{\text{quad}}$, in terms of those at $x = 0$, $\mathcal{R}_0^{\text{lin}}$ and $\mathcal{R}_0^{\text{quad}}$. In practice, however, an approximation has to be made. Since the ABH equation is singular at the origin, a small length imperfection l needs to be assumed and replace Y_0 , $\mathcal{R}_0^{\text{lin}}$ and $\mathcal{R}_0^{\text{quad}}$ with Y_l , $\mathcal{R}_l^{\text{lin}}$ and $\mathcal{R}_l^{\text{quad}}$.

The reflection coefficient for the linear case at the entrance of the ABH is provided in [20,23],

$$\mathcal{R}_L^{\text{lin}} = \frac{1 + \mathcal{R}_l^{\text{lin}} + \frac{1}{ik_0 L} (\alpha_+ + \mathcal{R}_l^{\text{lin}} \alpha_-)}{1 + \mathcal{R}_l^{\text{lin}} - \frac{1}{ik_0 L} (\alpha_+ - \mathcal{R}_l^{\text{lin}} \alpha_-)} e^{-i2k_0 L}, \quad (42)$$

with the reflection coefficient at $x = -l$, $\mathcal{R}_l^{\text{lin}}$, being

$$\mathcal{R}_l^{\text{lin}} = -\frac{\alpha_+ + ik_0 Z_0 Y_l}{\alpha_- + ik_0 Z_0 Y_l} \left(\frac{l}{L} \right)^{\alpha_+ - \alpha_-}. \quad (43)$$

If we want to consider a rigid end wall for the ABH we simply need to take $Y_l = 0$ in (43).

Likewise, the reflection coefficient for the quadratic case becomes [23],

$$\mathcal{R}_L^{\text{quad}} = \frac{\mathcal{R}_l^{\text{quad}} e^{ik_0 L} - e^{-ik_0 L} (1 + 2ik_0 L)}{\mathcal{R}_l^{\text{quad}} e^{ik_0 L} (1 - 2ik_0 L) - e^{-ik_0 L}}, \quad (44)$$

with

$$\mathcal{R}_l^{\text{quad}} = \frac{Y_l + i \frac{k_0 L^2}{l} \left(Y_l + \frac{l^2}{Z_0 l^2} \right)}{Y_l - i \frac{k_0 L^2}{l} \left(Y_l - \frac{l^2}{Z_0 l^2} \right)} e^{-2i \frac{k_0 L^2}{l}}. \quad (45)$$

Again we take $Y_l = 0$ in (45) for a rigid ABH end wall.

In what concerns the expression for the reflection coefficient at $x = -L$, when using the TMM approach, it can be easily obtained from (26). For a rigid termination $B_N = -A_N$, from which follows,

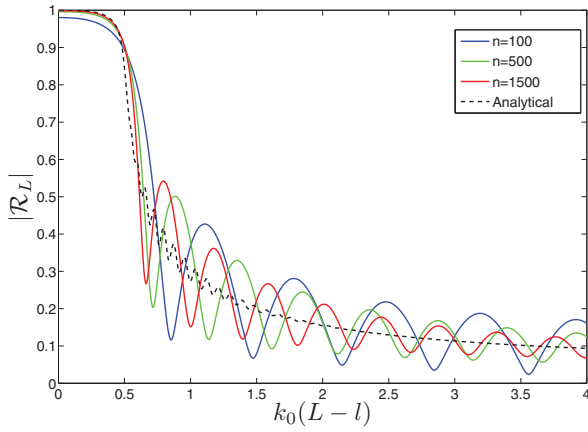
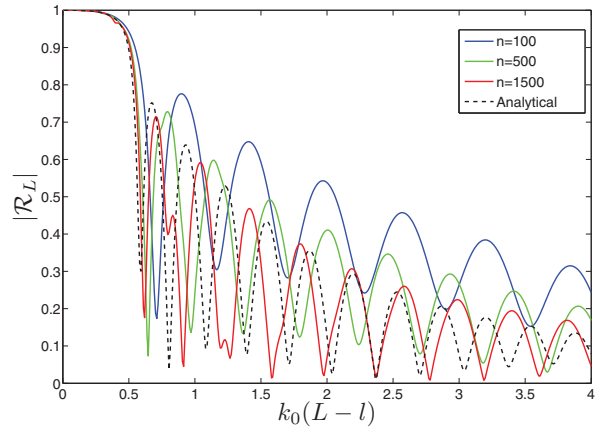
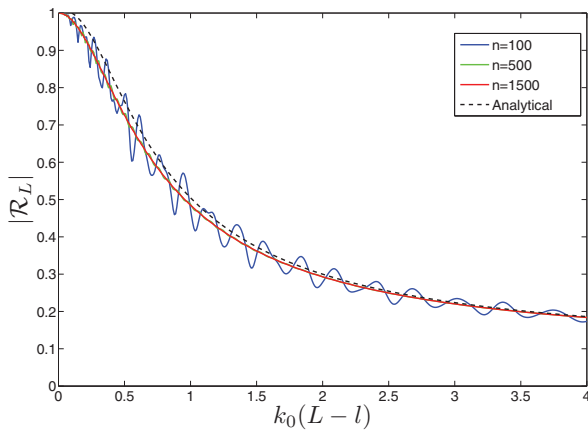
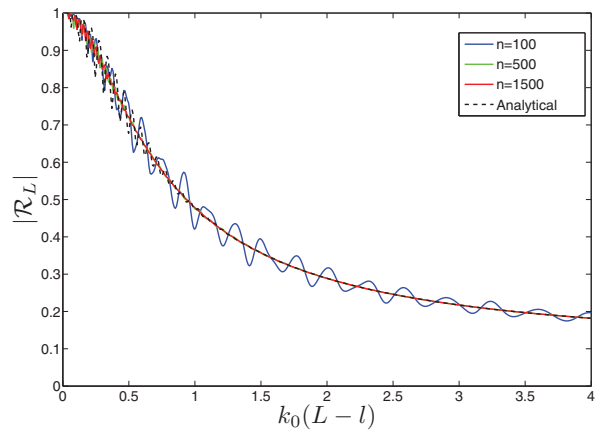
$$\mathcal{R}_L^{\text{TMM}} = \frac{\tilde{T}_{0,N}^{21} - \tilde{T}_{0,N}^{22}}{\tilde{T}_{0,N}^{11} - \tilde{T}_{0,N}^{12}}. \quad (46)$$

Our goal in the next subsection would be to check, for a few representative cases, whether the TMM reflection coefficient $\mathcal{R}_L^{\text{TMM}}$ in (46) tends to the exact ones, $\mathcal{R}_L^{\text{lin}}$ in (42) and $\mathcal{R}_L^{\text{quad}}$ in (44), when increasing the number of layers in the metafluid.

5.2. Simulations

For the simulations we consider a cylindrical duct with radius $R = 0.23$ m and length $L = 0.5$ m for both, the linear and quadratic cases. The cutoff frequency of the tube is $f_c = 1.84c_0/2\pi R = 445$ Hz, where $c_0 = 340$ m/s is the speed of sound. Damping has been introduced taking a complex speed of sound $c = c_0(1 + 0.05i)$ (see [20,23]).

In Figs. 4(a) and (b), we show the analytical and TMM reflection coefficients of the linear ABH when varying the number of TMM layers. Fig. 4(a) presents the values of $\mathcal{R}_L^{\text{lin}}$ and those of $\mathcal{R}_L^{\text{TMM}}$, respectively for 100, 500 and 1500 layers. As observed, the TMM results tend to the analytical one, although very slowly because the TMM matrices essentially consist of trigonometric functions. The critical value for the smoothness condition (19) occurs for $k_0 L = 1$ so the ABH only performs well (low reflection coefficient) well passed this value. The reflection coefficients in Fig. 4(a) have been computed

(a) Linear ($m = 1$), $L = 0.5$ m, $l = 5 \times 10^{-15}$ m(b) Linear ($m = 1$), $L = 0.5$ m, $l = 5 \times 10^{-5}$ m(c) Quadratic ($m = 2$), $L = 0.5$ m, $l = 5 \times 10^{-5}$ m(d) Quadratic ($m = 2$), $L = 0.5$ m, $l = 1 \times 10^{-2}$ m**Fig. 4.** Analytical and TMM reflection coefficients for the linear (a)-(b) and quadratic (c)-(d) ABHs.

assuming a very small imperfection of $l = 5 \times 10^{-15}$ m, for which the analytical solution barely oscillates. A higher value of $l = 5 \times 10^{-5}$ m is employed in Fig. 4(b). The results recognize a point already identified in [23]: the performance of the linear ABH is extremely sensitive to the value of the imperfection l . While almost negligible from a manufacturing point of view, an imperfection of $l = 5 \times 10^{-5}$ m produces a clear deterioration of the linear ABH performance.

The results for the quadratic ABH are exhibited in Figs. 4(c) and (d). Fig. 4(c) shows that the TMM solutions tend to the analytical one much faster for the second order ABH than for the linear one. Moreover, the quadratic ABH is more robust with respect to the imperfection length l than the linear one; the same value of $l = 5 \times 10^{-5}$ m in Fig. 4(b) has been used for Fig. 4(c) but $\mathcal{R}_L^{\text{quad}}$ does not fluctuate. It is necessary to increase the imperfection three orders of magnitude, up to $l = 1 \times 10^{-2}$ m (see Fig. 4(d)), to detect some oscillations in $\mathcal{R}_L^{\text{quad}}$. Likewise, note that for the quadratic case the critical value for the smoothness condition is $k_0 L = 2$, so a larger length L than for the linear case would be necessary to attain a low reflection coefficient, for the same fixed frequency.

6. Conclusions

In this work we wondered about the consistency of employing the transfer matrix method (TMM) to simulate the performance of acoustic black holes (ABHs) in duct terminations. In particular, we were interested in knowing whether the TMM solution to a discrete ABH termination, made of rings and cavities with a power-law decreasing inner radius, would formally tend to the solution of the ABH governing equation described in Mironov and Pislyakov original work. To begin with, we have shown that the discrete ABH can be interpreted as the practical realization of a metafluid whose density increases as it approaches the duct end wall. This connection has strongly simplified demonstrating consistency for the TMM and also opens the door to future ABH designs that could match the metafluid governing equation.

The next step has precisely consisted in transforming the metafluid equation into a Helmholtz equation with varying wavenumber. This has strongly simplified the application of the TMM because all that is needed is to discretize the metafluid into layers of constant characteristic impedance, with planar acoustic waves propagating in them. This provides a notably simple form of the transfer matrices. To demonstrate consistency, we have then discretized the Helmholtz equation with a second order finite difference scheme and substitute the TMM solution in it. Expanding the entries of all matrices involved in the resulting expression in Taylor series and making use of the ABH smoothness condition, we have proved that the differential Helmholtz equation is satisfied when the space step tends to zero. The same holds for the original ABH equation, when reverting the change of variables from the Helmholtz equation.

Finally, we have presented some numerical simulations for the reflection coefficients of linear and quadratic ABHs. The simulations reveal the TMM solutions tend to the analytical one, though very slowly for the linear ABHs, which are also especially sensitive to the ABH imperfection length. Conversely, quadratic ABHs are less affected by imperfections and approach faster the analytical solution. Nonetheless, for a fixed frequency quadratic ABHs require longer lengths than linear ABHs to achieve appropriate impedance matching.

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